

EduVision – Project Documentation

Higher Education Performance Dashboard | Final Project Documentation

1. Project Overview

EduVision – Higher Education Performance Dashboard is a data analytics and visualization project designed to compare universities, research performance, student characteristics, and country-level higher-education indicators. The project integrates higher-education ranking and reference datasets, prepares an analysis-ready dataset, derives KPI fields, and presents the results through an interactive Power BI dashboard.

2. Project Objectives

- Integrate university ranking and higher-education datasets into a consistent analytical dataset.
- Prepare reliable university-level and country-level metrics for dashboard analysis.
- Provide interactive views for university overview, research analytics, student analytics, and country comparison.
- Create reusable Python scripts for data collection, preparation, cleaning, validation, and KPI generation.
- Deliver a portfolio-ready Power BI dashboard with supporting documentation.

3. Dataset Sources

Source / File	Purpose
QS 2025.csv	University ranking and performance indicators.
THE 2025.csv	University ranking and higher-education performance indicators.
Education.csv / reference datasets	Reference structure and additional education-related fields used to align the analytical model.
Processed project datasets	Cleaned, integrated and KPI-enriched data used for dashboard development.

4. Data Processing Workflow

- Collect and inspect source datasets.
- Standardize university and country fields.
- Integrate ranking and reference information using appropriate university/year matching logic.
- Remove or resolve duplicate and inconsistent records.
- Validate column types, missing values, ranges, and key fields.
- Generate derived KPI fields and country-level aggregates.
- Export the final analysis-ready dataset for Power BI.
- Connect the final dataset to the dashboard and validate calculations.

5. Project Structure

/scripts – Python scripts for data collection, cleaning, and KPI generation.

/data/raw – Original/reference input datasets.

/data/processed – Cleaned datasets used for analysis.

/output – Final exported datasets and analytical outputs.

/dashboard – Power BI dashboard and dashboard storyboard.

/docs – Project documentation, KPI definitions, dashboard guide, and methodology.

6. Dashboard Suite

- **University Overview:** high-level university performance, ranking and institutional indicators.
- **Research Analytics:** research output, citations, reputation and research productivity indicators.
- **Student Analytics:** student population, international students, faculty and student-related indicators.
- **Country Comparison:** country-level ranking, averages, university counts and comparative performance.

7. Technology Stack

- Python and Pandas for data preparation and transformation.
- Power BI for interactive dashboard development and visualization.
- CSV/XLSX files for data exchange and analytical outputs.
- Git and GitHub for version control and project delivery.

8. Validation and Quality Assurance

- Validate KPI calculations against source and derived fields.
- Check ranking calculations and ordering.
- Check duplicate records and university uniqueness where applicable.
- Check dashboard filters, navigation, and cross-page interactions.
- Check that displayed metrics respond correctly to selected filters.

9. Deliverables

- GitHub repository containing source code, data structure, dashboard, and documentation.
- EduVision Power BI dashboard workbook.
- Final project documentation.
- KPI definitions.
- Dashboard user guide.
- Education analytics methodology.

10. Reproducibility Note

The repository is organized so that a reviewer can understand the path from raw data to processed data, KPI generation, and dashboard delivery. Temporary or duplicate reference files should be removed before final submission unless they are required to reproduce the final dataset.